
Chapter 8

Harmonic Progression

Real-Life Story (Why Harmonic Progression?)

Suppose the time taken to complete a task decreases as the number of workers increases.

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

Such sequences are connected with reciprocal relationships and are called **Harmonic Progressions (HP)**.

HP appears in physics, engineering, electrical circuits, speed problems, and wave calculations.

1 Harmonic Progression

Overview

A **Harmonic Progression (HP)** is a sequence in which the reciprocals of the terms form an Arithmetic Progression (AP).

We will learn:

- Definition of HP
- Relationship between HP and AP
- Harmonic Mean
- Properties of HP
- Applications of HP

1.1 Definition

A sequence

$$a_1, a_2, a_3, \dots$$

is called a Harmonic Progression if:

$$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$$

forms an Arithmetic Progression.

Important Idea

If the reciprocals of the terms form an AP, then the sequence is called an HP.

Example:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

Reciprocals:

$$1, 2, 3, 4, \dots$$

which is an AP.

2 General Form of HP

If

$$a, a + d, a + 2d, \dots$$

is an AP, then the corresponding HP is:

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots$$

3 Types of Harmonic Progression

1. Increasing HP

Example:

$$\frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \dots$$

2. Decreasing HP

Example:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

4 Harmonic Mean (HM)

If a, H, b are in HP, then:

$$H = \frac{2ab}{a+b}$$

Solved Example

Find the harmonic mean between 4 and 12.

Solution

Using:

$$\begin{aligned}H &= \frac{2ab}{a+b} \\H &= \frac{2(4)(12)}{4+12} \\&= \frac{96}{16} \\&= 6\end{aligned}$$

Hence,

$$\boxed{6}$$

5 Relationship Between AM, GM and HM

For two positive numbers:

$$AM \geq GM \geq HM$$

where:

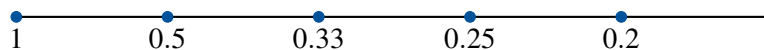
$$AM = \frac{a+b}{2}$$

$$GM = \sqrt{ab}$$

$$HM = \frac{2ab}{a+b}$$

6 Visual Understanding

Terms decrease harmonically



7 Properties of HP

- Reciprocals form an Arithmetic Progression.
- HP generally decreases slowly.
- Harmonic Mean is always less than Arithmetic Mean.
- Used in average speed and work problems.

8 Applications of HP

- Average speed calculations
- Electrical resistance problems
- Parallel circuits
- Physics and wave mechanics
- Work and time problems
- Engineering calculations

9 Solved Questions

Solved Question 1

Determine whether the sequence

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

is an HP.

Solution

Taking reciprocals:

$$1, 2, 3, 4, \dots$$

This forms an AP.

Hence, the given sequence is an HP.

Solved Question 2

Find the harmonic mean between 6 and 18.

Solution

Using:

$$H = \frac{2ab}{a+b}$$

$$\begin{aligned}
 H &= \frac{2(6)(18)}{6+18} \\
 &= \frac{216}{24} \\
 &= 9
 \end{aligned}$$

Hence,

$$\boxed{9}$$

Solved Question 3

Find the reciprocals of:

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$$

Solution

Reciprocals are:

$$2, 4, 6, \dots$$

which forms an AP.

Hence, the sequence is an HP.

Solved Question 4

Find the third term of the HP:

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

Solution

The third term is:

$$\boxed{\frac{1}{3}}$$

Solved Question 5

If 3, H, 27 are in HP, find H.

Solution

Using:

$$\begin{aligned}
 H &= \frac{2ab}{a+b} \\
 H &= \frac{2(3)(27)}{3+27}
 \end{aligned}$$

$$= \frac{162}{30}$$

$$= \frac{27}{5}$$

Hence,

$$\boxed{\frac{27}{5}}$$

Solved Question 6

Find the harmonic mean between 8 and 24.

Solution

$$H = \frac{2(8)(24)}{8 + 24}$$

$$= \frac{384}{32}$$

$$= 12$$

Hence,

$$\boxed{12}$$

Solved Question 7

Show that

$$\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$$

is an HP.

Solution

Reciprocals:

$$3, 4, 5$$

which forms an AP.

Hence, the sequence is an HP.

Solved Question 8

Find HM of 10 and 15.

Solution

$$H = \frac{2(10)(15)}{10 + 15}$$

$$= \frac{300}{25}$$

$$= 12$$

Hence,

$$\boxed{12}$$

Solved Question 9

Find the reciprocal sequence of:

$$2, 4, 6, 8, \dots$$

Solution

Reciprocal sequence:

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \dots$$

which is an HP.

Solved Question 10

Find the harmonic mean between 5 and 20.

Solution

$$H = \frac{2(5)(20)}{5 + 20}$$

$$= \frac{200}{25}$$

$$= 8$$

Hence,

$$\boxed{8}$$

10 Practice Questions

Basic Level

1. Determine whether:

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

is an HP.

2. Find the harmonic mean between 4 and 8.

3. Find reciprocals of:

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{6}$$

4. Find HM between 9 and 18.

5. Show that:

$$\frac{1}{5}, \frac{1}{6}, \frac{1}{7}$$

is an HP.

Advanced Level

6. Find HM between 12 and 30.

7. If 5, H, 45 are in HP, find H.

8. Prove:

$$AM \geq HM$$

9. Find reciprocal sequence of:

$$3, 6, 9, \dots$$

10. Find HM between 16 and 48.

11 Quick Formula Sheet

Concept	Formula	Meaning
Harmonic Mean	$H = \frac{2ab}{a+b}$	Middle harmonic term
Arithmetic Mean	$AM = \frac{a+b}{2}$	Average value
Geometric Mean	$GM = \sqrt{ab}$	Middle geometric term
Relation	$AM \geq GM \geq HM$	Mean inequality

Galileo Galilei

“Mathematics is the language with which God has written the universe.”